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①

Given,

$$y = \sin^{-1} x$$

differentiating w.r. to  $x$ , we get

$$y_1 = \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow y_1 (\sqrt{1-x^2}) = 1$$

$$\Rightarrow y_1^2 (1-x^2) = 1$$

again, differentiating w.r. to  $x$ , we get

$$\Rightarrow 2y_1 y_2 (1-x^2) + y_1^2 (0-2x) = 0$$

$$\Rightarrow (1-x^2) y_2 \cdot 2y_1 - 2xy_1 = 0$$

$$\Rightarrow (1-x^2) y_2 - xy_1 = 0$$

① From ①;

$$(1-x^2)y_2' - xy_1' = 0 \quad \text{--- Leibnitz's Theorem} \quad \text{①}$$

First part,  $(1-x^2)y_2$

let,

$$u = y_2$$

$$v = (1-x^2)$$

$$u_1 = y_3$$

$$v_1 = -2x$$

$$0 = u_1 v - u v_1 = y_3(1-x^2) - y_2(-2x)$$

$$y_3 = y_5$$

$$v_3 = 0$$

$$\underline{u_n = y_{n+2}}$$

$$u_n = y_{n+2}$$

by Leibnitz's Theorem,

$$(uv)_n = u_n v + n u_{n-1} v_1 + n u_{n-2} v_2 + \dots + u v_n$$

$$\{(1-x^2)y_2\}_n = y_{n+2}(1-x^2) + n y_{n+1}(-2x) + \frac{n(n-1)}{2} y_n \cdot (-2) + 0$$

$$= (1-x^2)y_{n+2} - 2nx y_{n+1} - n(n-1)y_n$$

Second part,  $-xy_1$

$$\text{let, } u = y_1$$

$$v = -x$$

$$\begin{pmatrix} \text{Leibnitz} \\ u_1 = y_2 \\ u_2 = y_3 \end{pmatrix}$$

$$v_1 = -1$$

$$v_2 = 0$$

$$\underline{\underline{u_n = y_{n+1}}}$$

by Leibnitz's Theorem,

$$(UV)_n = U_n V + n C_1 U_{n-1} V_1 + n C_2 U_{n-2} V_2 + \dots + U_n V_n$$

$$\{-xy\}_n = y_{n+1}(-x) + n y_n (-1) + 0$$

$$= -x y_{n+1} - n y_n$$

Thus from equation (1) is

$$(1-x^2) y_{n+2} - 2nx y_{n+1} - (n(n-1)y_n - x y_{n+1} - n y_n) = 0$$

$$\Rightarrow (1-x^2) y_{n+2} - (2n+1)x y_{n+1} - (n(n-1) - n) y_n = 0$$

$$\Rightarrow (1-x^2) y_{n+2} - (2n+1)x y_{n+1} - n^2 y_n = 0$$

(showed)

④ ① Given,  $y = \tan^{-1} x$

let's differentiate with the respect to  $x$ ,

$$y_1 = \frac{1}{1+x^2}$$

$$\Rightarrow y_1 (1+x^2) = 1$$

(showed)

① From equation ①,

$$(1+x^2) y_1 = 1 \quad \text{--- (1)}$$

$$\Rightarrow (1+x^2) y_1 - 1 = 0$$

part

Now differentiate equation (1) with the help of Leibnitz's theorem,

|                         |                    |                                      |
|-------------------------|--------------------|--------------------------------------|
| let, $u = y_1$          | $v = (1+x^2)^{-1}$ | $\frac{d}{dx} (-1)$                  |
| $\Rightarrow u_1 = y_2$ | $v_1 = 2x$         | $= 0$                                |
| $\Rightarrow u_2 = y_3$ | $v_2 = 2$          | $\therefore nC_1 = n$                |
| $\Rightarrow u_3 = y_4$ | $v_3 = 0$          | $\therefore nC_2 = \frac{n(n-1)}{2}$ |
| $\vdots$                | $\vdots$           | $\vdots$                             |
| $u_n = y_{n+1}$         | $v_n = 0$          |                                      |

by Leibnitz's theorem,

$$(uv)_n = u_n v + nC_1 u_{n-1} v_1 + nC_2 u_{n-2} v_2 + \dots + u_1 v_n$$

$$\{(1+x^2) y_1\}_n = y_{n+1} (1+x^2) + n y_n \cdot 2x + \frac{n(n-1)}{2} \cdot y_{n-1} \cdot 2 + 0$$

$$= (1+x^2) y_{n+1} + 2nx y_n + n(n-1) y_{n-1} \quad \text{--- (2)}$$

$\therefore$  Thus from equation (2)

$$(1+x^2) y_{n+1} + 2nx y_n + n(n-1) y_{n-1} = 0$$

(showed)



First part,  $(1+x^2)y_2$

Leibnitz's theorem

$$\begin{array}{l|l} \text{let } u = y_2 & v = 1+x^2 \\ u_1 = y_3 & v_1 = 2x \\ u_2 = y_4 & v_2 = 2 \\ u_3 = y_5 & v_3 = 0 + 2(1-x^2) = 2-2x^2 \\ \vdots & \vdots \\ u_n = y_{n+2} & \end{array}$$

(2) Leibnitz's theorem

$$0 = n!v^{(n)} + (n-1)!v^{(n-1)}u_1 + (n-2)!v^{(n-2)}u_2 + \dots + u_nv^{(n)}$$

$\therefore$  by Leibnitz's theorem,

$$0 = n!(1+x^2)u_n + (n-1)!v^{(n-1)}u_1 + (n-2)!v^{(n-2)}u_2 + \dots + u_nv^{(n)}$$

$$(uv)_n = u_nv + nC_1 u_{n-1}v_1 + nC_2 u_{n-2}v_2 + \dots + u_nv_n$$

$$\begin{aligned} \therefore \{(1+x^2)y_2\}_n &= (y_{n+2}(1+x^2) + n \cdot y_{n+1} \cdot 2x + \frac{n(n-1)}{2} \cdot y_n \cdot 2 + 0 \\ &= (1+x^2)y_{n+2} + 2nx y_{n+1} + n(n-1)y_n \end{aligned}$$

$$v^{(n)}(x) = 0 \quad \text{if } n \geq 3 \quad \text{--- (1) (2)}$$

Second part,  $(2x+1)y_1$

$$\begin{array}{l|l} \text{let } u = y_1 & v = (2x+1) \\ u_1 = y_2 & v_1 = 2 \\ u_2 = y_3 & v_2 = 0 \\ \vdots & \vdots \\ u_n = y_{n+1} & \end{array}$$

by Leibnitz's theorem,

$$(uv)_n = u_n v + n C_1 u_{n-1} v_1 + n C_2 u_{n-2} v_2 + \dots + u v_n$$

$$\begin{aligned} \{(2x-1)y\}_n &= y_{n+1}(2x-1) + n \cdot y_n \cdot 2 + 0 \\ &= (2x-1)y_{n+1} + 2n y_n \end{aligned}$$

Thus from equation (1)

$$(1+x^2)y_{n+2} + 2nx y_{n+1} + n(n-1)y_n + (2x-1)y_{n+1} + 2ny_n = 0$$

$$\Rightarrow (1+x^2)y_{n+2} + (2nx + 2x - 1)y_{n+1} + n(n+1)y_n = 0$$

$$0 + 2 \cdot \frac{(1-n)n}{2} + x \cdot (1+n) + (x-1) = 0 \quad (\text{shown})$$

$$n(n-1) + (1+n)x + (x-1) = 0$$

16) ① Given,  $y = (\sin^{-1} x)^2$

Differentiating w.r.t.  $x$ , we get,

$$y_1 = 2 \sin^{-1} x \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow y_1 (\sqrt{1-x^2}) = 2 \sin^{-1} x$$

$$\Rightarrow y_1^2 (1-x^2) = (2 \sin^{-1} x)^2$$

$$\Rightarrow (1-x^2)y_1^2 = 4y$$



Again differentiating w.r.t.  $x$ , we get,  $(1-x^2) \cdot 2y_1 y_2 + y_1^2 (0-2x) = 4 \cdot 1 \cdot y_1$

$$\Rightarrow (1-x^2) y_2 \cdot 2y_1 - x y_1 \cdot 2y_1 - 2 \cdot 2y_1 = 0$$

$$\Rightarrow (1-x^2) y_2 - x y_1 - 2 = 0$$

$$1-x^2 = v$$

$$0 = -xv$$

showed.

② From ques ①,

$$(1-x^2) y_2 - x y_1 - 2 = 0 \quad \text{--- ①}$$

now differentiating equation ①  $n$  times with the help of Leibnitz's theorem,  $y_1^{(n)} + n y_1^{(n-1)} + \dots + n y_1 = n(y_1)$

First part,  $(1-x^2) y_2^{(n)} + (x^2) y_2^{(n-1)} = n y_2^{(n-1)}$

$$\text{let, } u = y_2$$

$$u_1 = y_3$$

$$u_2 = y_4$$

$$u_3 = y_5$$

$$\vdots$$

$$v = 1-x^2$$

$$v_1 = -2x$$

$$v_2 = -2$$

$$v_3 = 0$$

Here,

$$n C_1 = n$$

$$n C_2 = \frac{n(n-1)}{2}$$

(1) notations must, with

$$0 = 0 - n C_1 u_1 v_1 + n C_2 u_2 v_2 - n C_3 u_3 v_3 + \dots + n C_n u_n v_n$$

by Leibnitz's theorem,  $(uv)^{(n)} = n C_1 u^{(n-1)} v + n C_2 u^{(n-2)} v_2 + \dots + n C_n u v_n$

$$(uv)^{(n)} = u_n v + n C_1 u_{n-1} v_1 + n C_2 u_{n-2} v_2 + \dots + n C_n u v_n$$

$$\{(1-x^2) y_2\}^{(n)} = y_{n+2} (1-x^2) + n y_{n+1} (-2x) + \frac{n(n-1)}{2} y_n (-2) + 0$$



$$(1-x^2)y_{n+2} - 2nx y_{n+1} - n(n-1)y_n = 0$$

$$L.H.S = (x^2 - 0) \cdot y_1 + 1 \cdot y_1 \cdot (x-1)$$

Second part,  $-xy_1$

$$\text{let, } u = y_1$$

$$u_1 = y_2$$

$$u_2 = y_3$$

$$\vdots$$

$$u_n = y_{n+1}$$

$$v = -x$$

$$v_1 = -1$$

$$v_2 = 0$$

$$\textcircled{1} \quad \underline{u_n = y_{n+1}}$$

$$0 = x - x^2 - x^2(x-1)$$

by Leibnitz's Theorem, (1) follows by differentiation

$$(uv)_n = u_n v + n_1 u_{n-1} v_1 + n_2 u_{n-2} v_2 + \dots + u_n v_n$$

$$\{ -xy_1 \}_n = y_{n+1} \cdot (-x) + n \cdot y_n \cdot (-1) + 0 \cdot y_{n-1} \cdot 0 + \dots + y_n \cdot 0$$

$$\text{Third part, } -2 \quad \left| \begin{array}{l} -xy_{n+1} - ny_n = v \\ x^2 - 1 = v \\ x - 1 = v \\ 0 = v \end{array} \right. \quad \left| \begin{array}{l} v = -x \\ v_1 = -1 \\ v_2 = 0 \\ v_3 = 0 \\ \vdots \end{array} \right.$$

Thus, from equation (1),

$$(1-x^2)y_{n+2} - 2nx y_{n+1} - n(n-1)y_n = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - (2n+1)x y_{n+1} - n(n-1+1)y_n = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - (2n+1)x y_{n+1} - n^2 y_n = 0$$

$$0 = (x-1) \cdot \frac{d}{dx} (1-x^2)y + (x^2-1) \cdot \frac{d}{dx} (2n+1)x y + (x-1) \cdot \frac{d}{dx} (n^2 y) \quad (\text{Showed})$$